JavaScript: Operators

An operator is a symbol that performs an operation on values or variables.
Operators allow us to:

- perform calculations
- compare values
- build logical conditions
- update variable values

Common operators

- Arithmetic
  + - * / %
- Comparision
  == === !! > <
- Logical
  && || !
- Asignment
  = += -=

Arithmetic Operators

Example:

```
let a = 10;
let b = 3;

console.log("Addition:", a + b);
console.log("Subtraction:", a - b);
console.log("Multiplication:", a * b);
console.log("Division:", a / b);
console.log("Remainder:", a % b);
```

The modulus operator `%` is commonly used when checking **even** or **odd** numbers.

Example:

```
let number = 10;

console.log(number % 2);
```

If the result is `0`, the number is even.

Comparison Operators

Operator	Meaning
==	Equal
===	Strict equal
!=	Not equal
>	Greater than
<	Less than

Difference Between == and ===

This is one of the most important things to understand in JavaScript.

== (Loose Equality)

`==` compares values after converting the type if necessary.

```
console.log(5 == "5");

// Output:
// true
// Because JavaScript converts "5" to a number.
```

=== (Strict Equality)

`===` compares both value and type.

```
console.log(5 === "5");

// Output:
// false
// Because:
// 5 → number
// "5" → string
```

> *Different types → result is **`false`**.*

Best Practice

Most developers prefer using **===** because it avoids unexpected type conversions.